DEAE: Diffusion-Enhanced Autoencoder Network for Unsupervised Nonlinear Hyperspectral Unmixing

Tongshu Wu, Fanqiang Kong, Dan Li, Yunsong Li, *Member, IEEE*, and Fei Ye

***Abstract*—Hyperspectral unmixing (HU) aims to decompose mixed pixels into their constituent spectral signatures and estimate their corresponding fractional abundances. Recently, the nonlinear spectral mixing model (NLMM) has advanced significantly and offered strong physical interpretability. However, the effective integration of physics-driven NLMM with data-driven deep learning (DL) approaches still remains a critical challenge. To address this, we propose a diffusion-enhanced autoencoder (DEAE), a novel unsupervised framework that innovatively incorporates the diffusion model (DM) into nonlinear HU. DEAE introduces the residual second-order attention mechanism to capture global spectral information, adaptively weighting informative bands while compressing redundant bands. Subsequently, we integrate the extended multilinear mixing model (EMLM) into the DM-enhanced decoder, which extracts latent features from the linear autoencoder's output and generates an enhanced reconstructed image while simultaneously estimating the transition probabilities of EMLM. Finally, a nonlinear decoder outputs the ultimate reconstructed image based on both the enhanced reconstructed image and transition probabilities. Experiments conducted on synthetic and three real-world datasets demonstrate the superior performance of DEAE compared to the state-of-the-art methods based on both LMM and NLMM.**

***Index Terms*—Autoencoder (AE), diffusion model (DM), hyperspectral image (HSI), unsupervised nonlinear unmixing.**

I. Introduction

HYPERSPECTRAL images (HSIs), owing to their abundant spectral and spatial information, have been extensively applied in environmental monitoring, ecological conservation, and mineral exploration [1]. However, the high spectral resolution of HSI sensors often compromises spatial resolution, resulting in mixed pixels where the spectral signature of a single pixel is a composite of all ground covers within the area. This complexity leads to inaccurate recognition of ground covers, hindering the development of hyperspectral remote sensing. To address the issue of mixed pixels, numerous hyperspectral unmixing (HU) methods have been proposed to decompose the mixed spectral curve into pure components (endmembers) and their corresponding proportions (abundances) [2].

HU methods are primarily developed based on spectral mixing models, which can be categorized into two types based on endmember mixing mechanisms: linear spectral mixing model (LMM) and nonlinear spectral mixing model (NLMM) [3]. LMM assumes that, before reaching the sensor, the incident light interacts with only one homogeneous material within an instantaneous field of view. Consequently, the observed spectrum constitutes a linear combination of endmember spectra. Owing to its straightforward spectrum combination approach and explicit physical significance, LMM has been widely used in the unmixing domain at present. Based on LMM, the vertex component analysis (VCA) [4], N-FINDR [5], pixel purity index (PPI) [6], and some methods are proposed to extract endmembers. When endmembers are known, the fully constrained least-squares (FCLS) [7] unmixing technique can be used to estimate fractional abundances. Then, nonnegative matrix factorization (NMF) [8] is proposed to jointly extract endmember and estimate abundance through matrix factorization constraints. The LMM has demonstrated satisfactory performance in scenarios where the ground covers are simple and flat. However, its results are less satisfactory when the Earth's surface features complex 3-D structures. In recent years, increasing attention has been paid to NLMM. A variety of NLMMs have been put forward to simulate the real situations of ground covers effectively. Among them, the Hapke model [9] is one of the earliest nonlinear models based on the radiative transfer equation [10]. This model can precisely describe the propagation of light in an intimate mixed medium, but it requires a large number of parameters for simulation. The bilinear mixing model (BMM) [11], which extends the LMM, accounts for secondary interactions of light among endmembers. BMM simplifies the propagation process while maintaining competitive accuracy. For instance, the Nascimento model [11], [12], the Fan model [13], the generalized bilinear model (GBM) [14], and the polynomial postnonlinear mixing model (PNMM) [15] are all typical BMMs. Li et al. [16] proposed the band-wise GBM (NU-BGBM) based on GBM, which calculates different parameters

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for different bands in HSI and is more suitable for the complex noise environment in real scenarios. These models only consider the second-order scattering effects between endmembers while ignoring higher order scattering of order three or more. Nevertheless, when handling complex scenarios, the higher order endmember interactions can also have a considerable impact on the observed spectrum [17].

The multilinear mixing model (MLM) [18] has been proposed to consider infinite-order endmember interactions. In detail, the MLM introduces transition probabilities $\mathbf{P}$ to describe higher order scattering interactions up to infinite orders while maintaining clear physical reasoning. The MLM model initially proposed by Heylen and Scheunders [18] is supervised, with endmember signatures serving as its prior knowledge. Subsequently, Wei et al. [19] proposed an unsupervised unmixing model based on MLM called MLMp that the endmembers are estimated jointly with the abundances and the transition probabilities. Li et al. [20] proposed graph-regularized MLM (G-MLM), which incorporates the Laplacian graph regularizers into MLM and utilizes the latent manifold structure of pixel spectra to enhance the estimation of abundance and nonlinear probability parameters. However, the MLM models mentioned above ignore the fact that light has different transition probabilities at different wavebands [21]. To better adapt to this property, an extended MLM (EMLM) [22] for the band-wise nonlinear unmixing is presented so that each band of a pixel will have its own transition probabilities to simulate further interactions. Yang and Wang [22] proposed the band-wise nonlinear spectral unmixing (BNLSU) method, which adds sparsity of abundances and smoothness of probability parameters to the objective function as regularization terms and then applies the alternating direction method of multipliers (ADMM) to solve this constrained optimization problem. Li et al. [23] proposed a robust unmixing method based on EMLM that adopts the $L_{2,1}$-norm-based loss to suppress the effect of noise. Afterward, Li et al. [24] proposed a coarse-to-fine method based on EMLM, which uses superpixel segmentation to obtain coarse HSI. The unmixing results of the coarse HSI are used as the prior knowledge for the fine HSI unmixing process.

With the development of computer technology, deep learning (DL) has been widely applied in HU [25], [26] [27], demonstrating superior computational efficiency and generalization [28]. The autoencoder (AE) framework serves as the baseline network for unmixing due to its exceptional feature extraction and data reconstruction capabilities [29]. Guo et al. [30] proposed an unmixing method based on the AE cascade utilizing the AE for the first time in HU to reduce the uncertainty of the model and the influence of noise. Su et al. [31] proposed DAEN, which pioneered a multilayer AE architecture for HU and established the precise mathematical relationship between AE and NMF. Palsson et al. [32] proposed convolutional neural network autoencoder unmixing (CNNAEU) to make full use of the spatial information by a convolutional neural network. Inspired by the perception mechanism, Gao et al. [33] proposed CyCU-Net that is the first study to transfer the high-level semantic information to the AE by relaxing the original pixel-level reconstruction assumption to cycle consistency dominated by the cascaded AEs. Rasti et al. [34] were inspired by the theory of deep image priors and proposed UnDIP, which proposes a collaborative framework utilizing a geometric endmember estimation as a prior in the unmixing process. Zhou and Rodrigues [35] proposed two ADMM-based HU networks, addressing both supervised and unsupervised scenarios. These networks were derived through ADMM algorithm unrolling, effectively integrating model-driven priors with data-driven learning techniques. Ghosh et al. [36] presented a transformer-based AE model, where a transformer architecture was applied to the latent space of the AE to enhance feature extraction and improve unmixing accuracy. The MSNet proposed by Yu et al. [37] reduces the dimensions of the original HSI and fuses it with the original HSI to learn the low-dimensional embeddings of the fused image and reconstruct data. Qi et al. [38], to fully utilize the spectral–spatial information through spectral partitioning, proposed MSSS-Net. Tao et al. [39] proposed A2SAN performing abundance estimation first and extracting endmembers after abundance estimation.

Recently, diffusion models (DMs) have demonstrated significant potential in HSI processing [40], leveraging their exceptional data fitting and reconstruction capabilities [41], [42]. DMs learn noise distributions via multistep denoising, enabling effective separation of noise from true signals. This mechanism preserves spectral features and spatial details even under high-noise conditions in HSI. DiffUn [43] proposes the first diffusion-based framework for semi-supervised HU. During the forward process, it extracts endmember priors from spectral libraries, while the reverse process samples both endmembers and corresponding abundances from the learned posterior distribution. Yu et al. [44] proposed UnmixDiff to generate abundance maps via DMs, where the synthesized abundance maps are linearly mixed with scene endmembers to produce realistic hyperspectral samples. However, unsupervised HU inherently involves extracting physically meaningful low-dimensional representations from complex high-dimensional data [33], prioritizing feature extraction over image reconstruction. Consequently, the application of DMs in unsupervised HU still remains in its early stages.

The DL-based methods introduced above mainly concentrate on LMM. In recent years, DL has also demonstrated significant potential in NLMMs. Initially, people focused their attention on the BMM and the Hapke model. Su et al. [45] presented a deep multitask bilinear unmixing (DMBU), which is an unsupervised unmixing approach based on BMM under the multimask learning framework. Rasti et al. [46] presented a convolutional encoder–decoder network based on the Hapke model (HapkeCNN). The convolutional operator helps to capture local homogeneity between abundances in intimate mixtures. With the development of NLMM, a variety of DL algorithms based on MLM have also emerged. Su et al. [47] proposed a deep AE-based augmented network (DAAN) for blind multilinear HU. At the encoding end of this network, spatial information is extracted through multiple 3-D convolutional layers, and the extracted features are used to calculate the transition probabilities $\mathbf{P}$ through a hierarchical representation. Fang et al. [48] proposed an AE-based network

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(MLM-AE) based on MLM, which is different from DAAN in that it calculates the transition probability matrix at the decoding end through multiple fully connected layers, interleaved with hyperbolic tangent activation functions. Li et al. [49] proposed an extended multilinear mixing model-inspired dual-stream network (EMLM-Net). This network is divided into two branches, which extract the spatial information of the original HSI and the coarse HSI. The weights are shared between the two branches, so as to better extract the nonlocal information. At the decoder, the transition probability $\mathbf{P}$ is calculated by ADMM to achieve a better performance.

The EMLM model accurately simulates light propagation by incorporating high-order scattering effects while designing band-wise transition probabilities $\mathbf{P}$ based on the spectral properties of endmembers. To combine the physical significance of EMLM with the representational capacity of generative models, we propose a diffusion-enhanced AE (DEAE), a novel framework that integrates DM into unsupervised nonlinear HU. Specifically, DEAE employs a diffusion-based decoder that begins with linear representations of reconstructed HSIs and progressively refines them by mapping transition probabilities $\mathbf{P}$ into the parameters of the DM. Through iterative computations of $\mathbf{P}$ via forward and reverse processes, DEAE provides a novel approach to simulate single-to-multiple scattering processes of light under a real environment. Moreover, estimating the band-wise $\mathbf{P}$ requires us to extract richer spectral information. We design a residual second-order channel attention (RSCA) in our model to assign greater weights to spectral bands rich in information and effectively compress redundant bands. The contributions can be summarized as follows.

1) DEAE represents the first integration of DMs with nonlinear HU. The proposed DEAE architecture combines the physical interpretability of EMLM with the powerful data-fitting capability of DMs, demonstrating superior robustness and unmixing accuracy.
2) DEAE introduces a novel mechanism for estimating transition probabilities $\mathbf{P}$, where nonlinear parameters are mapped into DM via forward–reverse diffusion processes. Accurate estimation of $\mathbf{P}$ enables DEAE to simulate higher order light interactions with greater precision.
3) DEAE incorporates a novel interspectral attention mechanism for HU, termed RSCA. This module extracts both nonlocal and local features in the spectral dimension by capturing the second-order information among the bands and promotes the nonlinear modeling ability of DEAE.

The rest of this article is organized as follows. Section II briefly reviews the relevant unmixing models. Section III explains the detail of the proposed network. In Section IV, we analyze the experimental results for both synthetic and real HSIs and compare with other state-of-the-art methods. Finally, Section V concludes this article.

II. Related Works

The EMLM is illuminated by LMM and extends the scope of interactions to all orders. Therefore, the related spectral mixing models involve LMM, MLM and EMLM.

A. Linear Spectral Mixing Model

The observed HSI is represented by $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_N) \in \mathbb{R}^{L\times N}$ containing L-bands and N pixels. Each pixel $\mathbf{x}_i = [x_{i,1}, x_{i,2}, \ldots, x_{i,L}]^{\mathrm{T}} \in \mathbb{R}^L, i = 1, 2, \ldots, N$, is constructed as a linear combination of endmembers, with their respective abundances as linear coefficients. The formula can be expressed as

$$\mathbf{x}_i = \sum_{j=1}^{k} \mathbf{e}_j a_{i,j} + \mathbf{n}_i = \mathbf{E}\mathbf{a}_i + \mathbf{n}_i \tag{1}$$

where $\mathbf{e}_j \in \mathbb{R}^L$ represents the spectral signature vector of the jth endmember, $\mathbf{E} = [\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_k] \in \mathbb{R}^{L\times k}$ means the endmember matrix, its each column means the spectra of one endmember, k is the number of the endmembers, $a_{i,j} \in \mathbb{R}$ is the jth endmember's abundance, $\mathbf{a}_i \in \mathbb{R}^k$ represents the abundance vector corresponding to the ith pixel, and $\mathbf{n}_i$ denotes the additive Gaussian noise vector. The constraints on the abundances are the abundance nonnegative constraint (ANC) and abundance sum-to-one (ASC) as follows:

$$\begin{cases} \forall i, j : a_{i,j} \geq 0 \quad \text{(ANC)} \\ \sum_{j=1}^{k} a_{i,j} = 1 \quad \text{(ASC)}. \end{cases} \tag{2}$$

For the whole HSI, (1) can be expressed as

$$\mathbf{X} = \mathbf{E}\mathbf{A} + \mathbf{n} \tag{3}$$

where $\mathbf{A} \in \mathbb{R}^{k\times N}$ is the abundance matrix and $\mathbf{n} \in \mathbb{R}^{L\times N}$ is the noise matrix.

B. Multilinear Mixing Model

However, LMM cannot deal with multiple reflections that often occur in many situations, and several authors have hence augmented the LMM with infinite terms named MLM. MLM includes all degrees of pure ground cover interactions and has a clear physical significance, based on the following assumptions. First, a light ray incoming from the source will interact with at least one material. Then, the ray will have a probability P_i of undergoing further interactions and a probability $(1 - P_i)$ of reaching the sensor, where $P_i < 1$ represents the transition probability in the pixel $\mathbf{x}_i$. Second, P_i is proportional to the endmember's abundance. Finally, when a light is scattered by endmember j, its intensity is correlated with the endmember's spectra $\mathbf{e}_j$. According to these assumptions, MLM can be represented as

$$\begin{aligned} \mathbf{x}_i &= (1 - P_i) \sum_{j=1}^{k} \mathbf{e}_j a_{i,j} + (1 - P_i) P_i \sum_{j=1}^{k} \sum_{r=1}^{k} \left(\mathbf{e}_j \odot \mathbf{e}_r\right) a_{i,j} a_{i,r} \\ &\quad + (1 - P_i) {P_i}^2 \sum_{j=1}^{k} \sum_{r=1}^{k} \sum_{h=1}^{k} \left(\mathbf{e}_j \odot \mathbf{e}_r \odot \mathbf{e}_h\right) a_{i,j} a_{i,r} a_{i,h} \\ &\quad + \cdots + \mathbf{n}_i \\ &= (1 - P_i)\, \mathbf{y}_i + (1 - P_i) P_i \mathbf{y}_i^{(2)} + (1 - P_i) {P_i}^2 \mathbf{y}_i^{(3)} \\ &\quad + \cdots + \mathbf{n}_i \\ &= (1 - P_i)\, \mathbf{y}_i + P_i \mathbf{y}_i \odot x_i + \mathbf{n}_i \end{aligned} \tag{4}$$

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where $\odot$ is the Hadamard product, or element-wise product, of two vectors; $\mathbf{y}_i = \sum_{j=1}^{k} e_j a_{i,j} \in \mathbb{R}^L$ denotes the linear combination of endmembers; and $\mathbf{y}_i^{(k)}$ means the kth Hadamard power of $\mathbf{y}_i$. By introducing the transition probability P, the MLM can include all orders of interactions in the ith pixel. Moreover, MLM can be reduced to LMM when $P = 0$. As P increases, higher order scatterings in scenarios will have a higher probability. The order of multiple scattering approaches infinity as P approaches one. In [18], the MLM is also well-defined for $P_i < 0$ even though the physical reasoning no longer holds for negative probability.

C. Extended Multilinear Mixing Model

In [22], the EMLM is derived by vectorizing the transition probabilities $\mathbf{P}_i = [P_{i,1}, P_{i,2}, \ldots, P_{i,L}]^{\mathrm{T}} \in \mathbb{R}^L$ to explain the band-wise change of nonlinearity. Specifically, each pixel in different bands will have different scalar transition probabilities. Thus, $P_{i,r}$ represents the transition probability corresponding to the rth band in the ith pixel. Based on (4), the EMLM can be represented as

$$\begin{aligned}\mathbf{x}_i &= (\mathbf{1}_L - \mathbf{P}_i) \odot \mathbf{y}_i + (\mathbf{1}_L - \mathbf{P}_i) \odot \mathbf{P}_i \odot \mathbf{y}_i^{(2)} \\ &\quad + (\mathbf{1}_L - \mathbf{P}_i) \odot \mathbf{P}_i^{(2)} \odot \mathbf{y}_i^{(3)} + \cdots + \mathbf{n}_i \\ &= (\mathbf{1}_L - \mathbf{P}_i) \odot \mathbf{y}_i + \mathbf{P}_i \odot \mathbf{y}_i \odot \mathbf{x}_i + \mathbf{n}_i \end{aligned} \tag{5}$$

where $\mathbf{1}_L \in \mathbb{R}^L$ is the all-ones vector. Then, we simplify the noise, and (5) can be derived as

$$\mathbf{x}_i = \frac{(\mathbf{1}_L - \mathbf{P}_i) \odot \mathbf{y}_i}{\mathbf{1}_L - \mathbf{P}_i \odot \mathbf{y}_i} + \mathbf{n}_i. \tag{6}$$

Based on (2), (3), and (6), the whole HSI $\mathbf{X} \in \mathbb{R}^{L \times N}$ can be represented as

$$\begin{aligned}\mathbf{X} &= (\mathbf{1} - \mathbf{P}) \odot \mathbf{Y} + (\mathbf{1} - \mathbf{P}) \odot \mathbf{P} \odot \mathbf{Y}^{(2)} \\ &\quad + (\mathbf{1} - \mathbf{P}) \odot \mathbf{P}^{(2)} \odot \mathbf{Y}^{(3)} + \cdots + \mathbf{n} \\ &= \frac{(\mathbf{1} - \mathbf{P}) \odot \mathbf{Y}}{\mathbf{1} - \mathbf{P} \odot \mathbf{Y}} + \mathbf{n} \\ \mathbf{A} &\geq 0 \quad \text{and } \mathbf{1}_k^{\mathrm{T}} \mathbf{A} = \mathbf{1}_N^{\mathrm{T}}, \quad \mathbf{P} < 1 \end{aligned} \tag{7}$$

where $\mathbf{1} \in \mathbb{R}^{L \times N}$ represents the all-ones matrix, $\mathbf{P} = [\mathbf{P}_1, \mathbf{P}_2, \ldots, \mathbf{P}_N] \in \mathbb{R}^{L \times N}$ is the global transition probabilities, and $\mathbf{Y} = \mathbf{EA}$ denotes the linear component of EMLM.

D. Diffusion Models

DMs, a class of generative models, have gained widespread applications in image generation [50], [51], image super-resolution [52], [53], natural language processing [54], [55], and other domains. Distinguished from conventional generative models, DMs progressively transmit information through latent states across multiple timesteps, forming a Markov chain of latent states. Specifically, each timestep in DMs only needs to model tiny changes, avoiding the complexity of single-step global modeling in traditional variational AEs while improving reconstruction accuracy. The denoising diffusion probabilistic model (DDPM) [56] is one of the most widely used DMs, which consists of a forward process and a reverse process. In the forward process, Gaussian noise is gradually added to the input data through a Markov chain until the data distribution approximates a standard normal distribution. The reverse process employs a neural network to progressively predict and remove noise, thereby reconstructing the original data from pure noise. In this article, we focus on the HU task and extend the DDPM framework to deeply explore latent features in HSI for the precision estimation of nonlinear parameters.

III. Proposed Method

In this section, we will elaborate on the proposed unsupervised unmixing network, detailing the RSCA module and the DEAE used to simulate the EMLM mechanism.

A. Network Overview

To better integrate the nonlinear mixing model with DL technology and fully utilize the spatial–spectral information of HSI along with the powerful data-fitting capability of AEs, a DEAE network named DEAE is presented. DEAE exploits the RSCA module to capture the correlation between channels and to enhance the feature channels that contain more information, while the abundant spectral information improves the accuracy of the estimated band-wise transition probabilities $\mathbf{P}$. Afterward, we extract spatial information and compress the HSI into its low-dimensional representation, obtaining the abundance vector.

In the decoder part, DEAE operates in three steps. First, the abundance vector is reconstructed into its corresponding linear representation according to (3). Then, the nonlinear reconstruction process is modeled via a DM to compute the transition probabilities $\mathbf{P}$. Finally, DEAE reconstructs the linear representation into its corresponding nonlinear representation from the linear representation and $\mathbf{P}$ based on (7). The decoder effectively captures the inherent relationship between the LMM and EMLM models. The overall framework is illustrated in Fig. 1.

B. Residual Second-Order Channel Attention

DEAE calculates independent $\mathbf{P}$ for different bands. However, some bands containing irrelevant information may be assigned high transition probabilities, introducing additional redundancy into the nonlinear reconstruction process and degrading the accuracy and quality of the reconstructed HSI. To address this, we employ RSCA that enables the network to focus more effectively on the critical spectral bands.

The channel attention mechanism currently employed for unmixing primarily utilizes global average pooling (GAP) on the input image, followed by fully connected layers or multilayer perceptrons (MLPs) for data-driven feature learning. Some methods also combine GAP with global max pooling (GMP) for processing [57], [58]. In contrast, RSCA introduces a more interpretable approach to feature extraction by calculating a covariance matrix across different spectral bands on the input image to assess interband correlations. First, when calculating the covariance matrix, each band is associated with every other band, fully capturing global information. Second, the covariance matrix effectively measures the correlation between bands, where higher correlations correspond to

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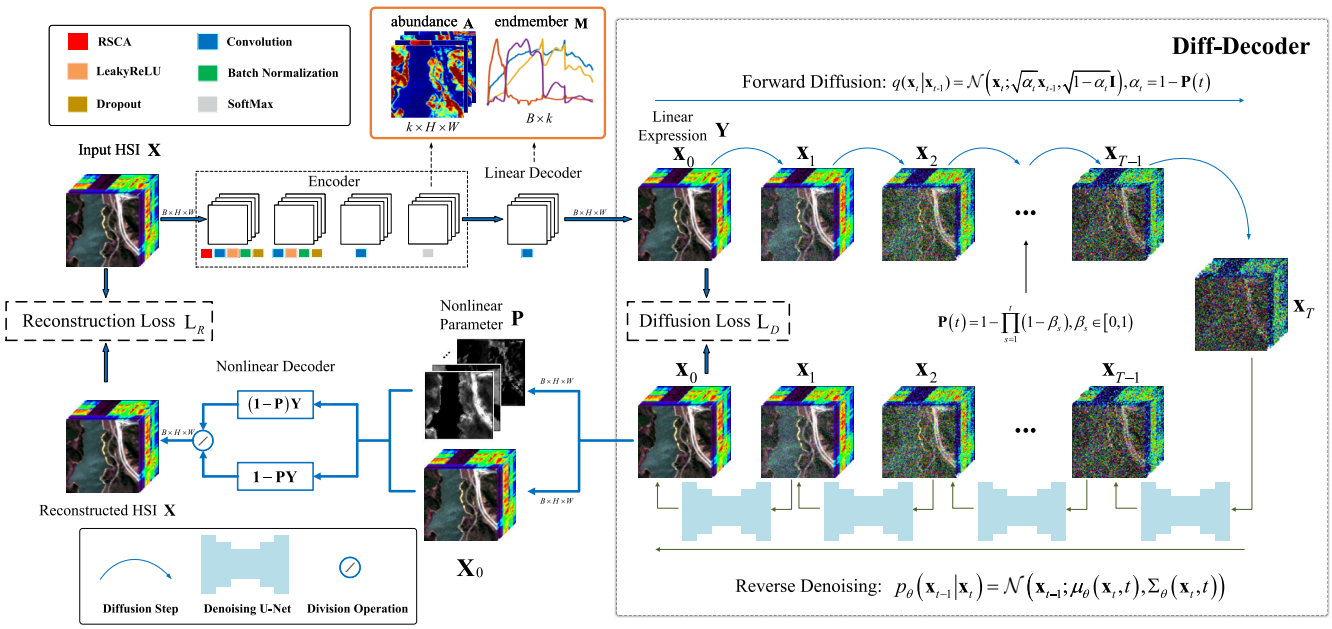


Fig. 1. Architecture of DEAE.

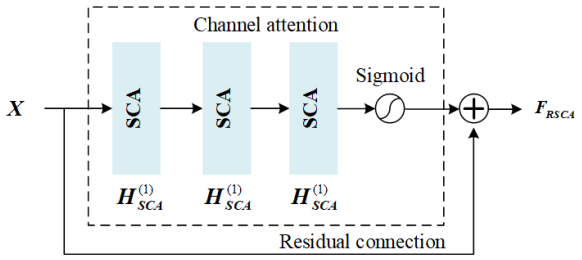


Fig. 2. Architecture of RSCA.

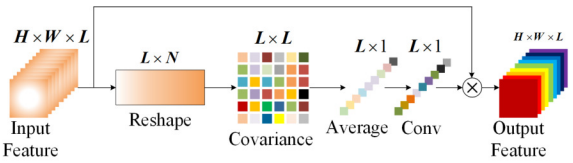


Fig. 3. Architecture of SCA module.

larger absolute values. Therefore, bands with high correlations to most other bands contain essential information, whereas those with lower correlations are redundant. Additionally, as a second-order matrix, the covariance matrix inherently provides nonlinear fitting capabilities, which enhance robustness and assist the subsequent network's unmixing tasks based on the NLMM.

As shown in Fig. 2, RSCA cascades three second-order channel attention (SCA) modules and connects them with residual connections. Suppose that the input HSI is $\mathbf{X} \in \mathbb{R}^{H\times W\times L}$, where H and W represent the spatial size and L is the number of bands. The function of RSCA can be expressed as

$$\begin{aligned}\mathbf{F}_{\mathrm{RSCA}} &= H_{\mathrm{RSCA}}(\mathbf{X}) \\ &= H^{(3)}_{\mathrm{SCA}}\left(H^{(2)}_{\mathrm{SCA}}\left(H^{(1)}_{\mathrm{SCA}}(\mathbf{X})\right)\right) + \alpha\mathbf{X}\end{aligned} \tag{8}$$

where $H^{(i)}_{\mathrm{SCA}}$ means the function of ith SCA module and α is the tradeoff weight in residual connections to balance the extracted feature and the input.

The detailed architecture of the SCA module is shown in Fig. 3. Assume that the input of the ith SCA module is $\mathbf{F}^{(i-1)}_{\mathrm{RSCA}} \in \mathbb{R}^{H\times W\times L}$. First, the 3-D input feature is reshaped and then transposed to form $\mathbf{F}_r \in \mathbb{R}^{L\times N}$, where $N = H \times W$ represents the spatial resolution, and each column corresponds to one pixel. Then, we calculate the covariance matrix of $\mathbf{F}_r$, which can be expressed as

$$\mathbf{F}_c = \frac{1}{N-1}\left(\mathbf{F}_r - \bar{\mathbf{F}}_r\right)\left(\mathbf{F}_r - \bar{\mathbf{F}}_r\right)^{\mathrm{T}} \tag{9}$$

$$\bar{\mathbf{F}}_r = \mu \cdot \mathbf{1}_N^{\mathrm{T}} \in \mathbb{R}^{L\times N} \tag{10}$$

where $\mathbf{F}_c \in \mathbb{R}^{L\times L}$ denotes the covariance matrix. $\mu \in \mathbb{R}^{L\times 1}$ is the mean column vector of $\mathbf{F}_r$, and $\bar{\mathbf{F}}_r$ repeats μ for N times in column. Then, we compute the average of each row in the matrix to determine the global correlation of each band. The formula of $\mathbf{F}_a$ can be expressed as

$$\mathbf{F}_a = \frac{1}{L}\mathbf{F}_c \cdot \mathbf{1}_L \tag{11}$$

where $\mathbf{F}_a \in \mathbb{R}^{L\times 1}$ denotes the average correlation between channels. Adjacent bands in HSI contain more relevant information. To effectively utilize this information, we designed a 1-D convolutional layer with a kernel size of 3, where the weights are continuously updated as the network iterates. This leverages the powerful data-fitting capability of neural networks to further extract deeper spectral information from adjacent bands. The formula of convolutional layer can be represented as

$$\mathbf{F}_{\mathrm{conv}} = \mathbf{W}_3 \odot \mathbf{F}_a \tag{12}$$

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where $\mathbf{W}_3$ denotes the weight corresponding to the 1-D convolutional layer with a kernel size of 3 and $\mathbf{F}_{\text{conv}}$ is the weight vector calculated by the SCA module. The output of the SCA module can be represented as

$$\begin{aligned}\mathrm{F}_{\mathrm{SCA}}^{(i)} &= H_{\mathrm{SCA}}^{(i)}\left(\mathrm{F}_{\mathrm{SCA}}^{(i-1)}\right)\\ &= \mathrm{F}_{\mathrm{SCA}}^{(i-1)} \odot \mathrm{F}_{\mathrm{conv}}\\ &= \mathrm{F}_{\mathrm{SCA}}^{(i-1)} \odot \mathbf{W}_3 \odot \left(\mathbf{F}_r - \bar{\mathbf{F}}_r\right)\left(\mathbf{F}_r - \bar{\mathbf{F}}_r\right)^{\mathrm{T}}\end{aligned} \tag{13}$$

where $H_{\mathrm{SCA}}(\cdot)$ denotes the function of SCA module.

C. Linear Encoder–Decoder

DEAE can be divided into two components: a linear encoder–decoder and nonlinear decoders. Assuming that the input of the linear encoder–decoder is $\mathbf{X} \in \mathbb{R}^{H\times W\times L}$, the encoder can be expressed as

$$\hat{\mathbf{A}} = H_E(\mathbf{X}) = \mathrm{Softmax}\left(\varepsilon\left(H_{\mathrm{RSCA}}(\mathbf{X})\right)\right) \tag{14}$$

where $\hat{\mathbf{A}} \in \mathbb{R}^{H\times W\times k}$ denotes the estimated abundance, k is the number of endmembers, and $H_E(\cdot)$ represents the function of encoder. $\varepsilon(\cdot)$ represents a feature extraction part where convolutional layers and LeakyReLU activations extract hierarchical features, while BN and dropout layers operate jointly to prevent overfitting and accelerate convergence. Afterward, the softmax layer is applied to the output of $\varepsilon(\cdot)$, thereby imposing ASC and ANC constraints on the output.

According to the EMLM model, we need to obtain the linear representation of the reconstructed image before the nonlinear decoder. A linear decoder utilizes a 1-D convolutional layer to process the linear transformation based on (3). The function of linear decoder $H_{D_L}(\cdot)$ can be expressed as follows:

$$\hat{\mathbf{Y}} = H_{D_L}\left(\hat{\mathbf{A}}\right) = \mathbf{W_L}^{(D_L)}\hat{\mathbf{A}} \tag{15}$$

where $\hat{\mathbf{Y}}$ is the linear representation of the reconstructed image and $\mathbf{W}_L^{(D_L)}$ denotes the weight matrix of 1-D convolutional layer with a kernel size of 1 in the linear component decoder. To reduce the training time, traditional algorithms for endmember extraction such as VCA [4] and N-FINDR [5] can be used to initialize $\mathbf{W}_L^{(D_L)}$. Based on the definition of the LMM, $\mathbf{W}_L^{(D_L)}$ can be interpreted as the endmember matrix. The extracted endmember can be expressed as

$$\mathbf{E} \leftarrow \mathbf{W}_L^{(D_L)}. \tag{16}$$

D. Diffusion-Enhanced Nonlinear Decoders

In the EMLM model, $\mathbf{P} \in \mathbb{R}^{L\times N}$ represents the likelihood of higher order interactions occurring in the propagation process for different bands of the endmembers. DEAE extends the forward–reverse diffusion processes through the Diff-Decoder design for accurate estimation of $\mathbf{P}$. Unlike traditional DDPM that relies solely on data distribution modeling, the proposed approach incorporates physical model constraints from EMLM to guide the generation process, establishing a physics-informed DM framework based on EMLM.

1) Forward Diffusion Process: We define the input of the Diff-Decoder as $\mathbf{x}_0 = \hat{\mathbf{y}}_i \in \mathbb{R}^L$, representing the ith pixel of the linear reconstructed image, and the linear reconstructed image after adding Gaussian noise t times is expressed as $\mathbf{x}_t$. In the forward process, Gaussian noise is added to $\mathbf{x}_0$ step by step over T time steps, which follows the Markov process. The conditional probability distribution of $\mathbf{x}_t$ given $\mathbf{x}_{t-1}$ can be expressed as

$$\mathbf{x_t} = \sqrt{1-\beta_t}\mathbf{x_{t-1}} + \sqrt{\beta_t}\epsilon_t,\ \epsilon_t \sim \mathcal{N}(0,\mathbf{I}) \tag{17}$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\mathbf{x}_t;\ \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right) \tag{18}$$

where $\mathcal{N}(\cdot)$ denotes the Gaussian distribution and β_t is the noise schedule parameters. Through iterative computation, the relationship between $\mathbf{x}_t$ and $\mathbf{x}_0$ can be derived as

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_0 + \sqrt{1-\alpha_t}\epsilon,\ \epsilon \sim \mathcal{N}(0,\mathbf{I}) \tag{19}$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_t;\ \sqrt{\alpha_t}\mathbf{x}_0, (1-\alpha_t)\mathbf{I}\right) \tag{20}$$

where $\alpha_t = \prod_{s=1}^{t}(1-\beta_s)$.

Before integrating the EMLM into the forward process, we first establish their mathematical consistency. From (5), the nonlinear term can be approximated as a superposition of linear terms and noise through the Taylor expansion. $y_i^{(n)}$ can be expressed as

$$\mathbf{y}_i^{(n)} = \mathbf{y}_i + \epsilon_i^{(n)},\ \epsilon_i^{(n)} \sim \mathcal{N}\left(0, \sigma_i^2\mathbf{I}\right). \tag{21}$$

Then, (5) can be obtained as

$$\begin{aligned}\mathbf{x}_i &= (1-\mathbf{P}_i)\mathbf{y}_i + (1-\mathbf{P}_i)\mathbf{P}_i\left(\mathbf{y}_i + \epsilon_i^{(2)}\right)\\ &\quad + (1-\mathbf{P}_i)\mathbf{P}_i^2\left(y_i + \epsilon_i^{(3)}\right) + \cdots + \eta\\ &= \left[(1-\mathbf{P}_i) + (1-\mathbf{P}_i)\mathbf{P}_i + \ldots\right]\mathbf{y}_i\\ &\quad + (1-\mathbf{P}_i)\mathbf{P}_i\epsilon_i^{(2)} + (1-\mathbf{P}_i)\mathbf{P}_i^2\epsilon_i^{(3)} + \cdots + \eta_i\\ &= \nu\mathbf{y}_i + \omega_{\text{total}}\end{aligned} \tag{22}$$

where ν can be expressed as

$$\nu = \sum_{n=1}^{\infty}(1-\mathbf{P_i})\mathbf{P_i}^{n-1}. \tag{23}$$

Combining all noise terms and assuming independent noise components with equal variance σ_n^2, we obtain

$$\omega_{\text{total}} \sim \mathcal{N}\left(0, \left(\sum_{n=2}^{\infty}(1-\mathbf{P}_i)^2\mathbf{P}_i^{2(n-1)}\sigma_n^2 + \sigma_\eta^2\right)I\right). \tag{24}$$

Equations (22)–(24) indicate that the randomness of photon multiple scattering causes higher order terms to deviate from linear superposition, manifesting as additive Gaussian noise. To establish the connection between EMLM and DMs, the transition probability can be expressed as

$$\mathbf{P}_i(t) = 1 - \prod_{s=1}^{t}(1-\beta_s) \tag{25}$$

$$\alpha_t = \prod_{s=1}^{t}(1-\beta_s) = 1 - \mathbf{P}_i(t). \tag{26}$$

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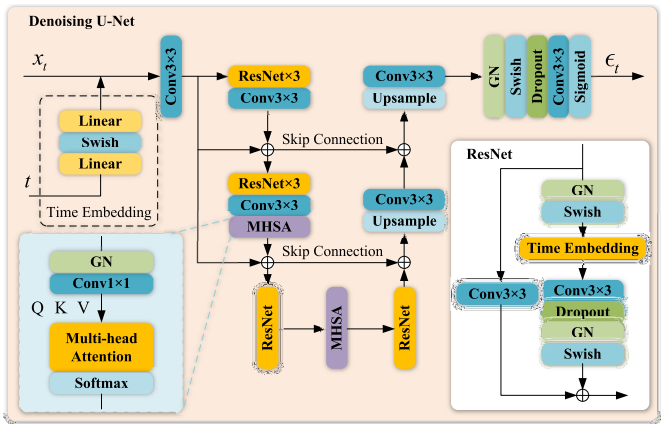


Fig. 4. Architecture of denoising U-Net.

Through the aforementioned parameterization, the series expansion of EMLM can be mapped to the forward process of the DM

$$\mathbf{x}_t = \sqrt{1-\mathbf{P}_i(t)}\mathbf{y}_i + \sqrt{\mathbf{P}_i(t)}\epsilon. \tag{27}$$

When $t = 0$, $\mathbf{P}_i(t) = 0$, the DM preserves the complete linear signal, and the spectral mixing model converges LMM. As t increases, $\mathbf{P}_i(t)$ increases with β_i, progressively activating higher order terms of EMLM, which corresponds to the rising noise proportion in DM. When $t = T$, $\mathbf{P}_i(t) = 1, \mathbf{x}_T \sim \mathcal{N}(0, 1)$ becomes a pure Gaussian noise. According to (5), EMLM asymptotically converges to the same noise distribution, thereby establishing the physical equivalence between the diffusion process and EMLM through identical noise dynamics.

2) Reverse Diffusion Process: The Diff-Decoder implements the reverse diffusion Markov chain that transforms the Gaussian-distributed $\mathbf{x}_T$ back to the linear reconstructed image $\mathbf{y}_i$ through T denoising steps. In the continuous limit where $T \to \infty$ and $\beta \to 0$, the reverse transition kernel $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ becomes Gaussian and can be expressed as

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t)) \tag{28}$$

where the reverse process is reparameterized through the auto-encoding variational Bayes approach [59] by estimating $\mu_\theta(\mathbf{x}_t, t)$ while fixing $\sum_\theta(\mathbf{x}_t, t) = \sigma_t^2\mathbf{I}$, where the variance σ_t^2 remains nontrainable. In practice, instead of directly predicting $\mu_\theta(\mathbf{x}_t, t)$, the model learns to estimate the noise component ϵ through a denoising U-Net architecture as specified in (19). Then, the parameterization of $\mu_\theta(\mathbf{x}_t, t)$ can be derived as

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{1-\beta_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\alpha_t}}\epsilon_\theta(x_t, t)\right). \tag{29}$$

The architecture of the denoising U-Net is illustrated in Fig. 4, where the inputs consist of the timestep and its corresponding HSI, while the output is the estimated noise. First, the timestep is processed by an MLP to generate a temporal feature vector. Subsequently, a 3×3 convolution layer maps the input to the initial feature space. In the downsampling stage, multiple ResNet modules are employed to progressively extract features while incorporating temporal embeddings. Notably, skip connections preserve hierarchical features from each layer. A multihead self-attention (MHSA) mechanism is introduced before the final downsampling layer to capture global dependencies. The MHSA module consists of a group normalization (GN) layer, a 1×1 convolutional layer, a multihead attention layer, and a softmax layer. The upsampling stage inversely utilizes the features from the downsampling path through skip connections. Finally, a sigmoid activation constrains the output range to $[0, 1]$. The loss function can be formulated as

$$\begin{aligned}\mathcal{L}_{\text{diff}} &= \mathbb{E}_{t,\epsilon}\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2 \\ &= \mathbb{E}_{t,\epsilon}\left\|\epsilon - \epsilon_\theta\left(\sqrt{\alpha_t}\mathbf{x}_0 + \sqrt{1-\alpha_t}\epsilon\right)\right\|^2.\end{aligned} \tag{30}$$

Algorithm 1 Pseudocode of the Proposed DEAE

Input: Original HSI $\mathbf{X}$, Endmember number k
Output: Endmembers $\mathbf{E}$, Abundances $\mathbf{A}$, Transition probabilities $\mathbf{P}$

Initialize $\mathbf{E}$ by VCA and k
for *Epochs* **do**
  $\mathbf{F}_{\text{RSCA}} \leftarrow H_{\text{RSCA}}(\mathbf{X})$
  $\hat{A} \leftarrow \text{Softmax}(\varepsilon(\mathbf{F}_{\text{RSCA}}))$
  $\hat{Y} \leftarrow \mathbf{W_L}^{(D_L)}\hat{A}$
  $\mathbf{E} \leftarrow \mathbf{W}_L^{(D_L)}$
  $\mathbf{y}_i \leftarrow \hat{Y}_{:,i}$
  while $t < T$ **do**
    $t \sim \text{Uniform}(1, \ldots, T)$
    $\mathbf{P}_i(t) = 1 - \prod_{s=1}^{t}(1-\beta_s), \beta_s \in [0, 1)$
    $q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\mathbf{P}(t)}\mathbf{x}_{t-1}, \sqrt{\mathbf{P}(t)}\mathbf{I})$
    $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t))$
  end
  $\hat{\mathbf{X}} \leftarrow \frac{(1-\mathbf{P})\odot\hat{X}_0}{1-\mathbf{P}\odot\hat{X}_0}$
end

E. Nonlinear Decoder

Following the Diff-Decoder's output of the estimated transition probabilities $\mathbf{P}$ and reconstructed HSI $\hat{\mathbf{X}}_0$, the nonlinear decoder shown in Fig. 1 reconstructs the nonlinear representation of HSI. Based on (7), the function of nonlinear decoder can be represented as

$$\hat{\mathbf{X}} = H_{ND}(\hat{\mathbf{X}}_0, \mathbf{P}) = \frac{(1-\mathbf{P})\odot\hat{\mathbf{X}}_0}{1-\mathbf{P}\odot\hat{\mathbf{X}}_0}. \tag{31}$$

In summary, the pseudocode of DEAE is shown in Algorithm 1.

F. Loss Function

In the design of the loss function, we introduced the commonly used the spectral angle distance (SAD) and the mean square error (mse) as loss functions of reconstructed HSI. SAD can effectively reduce spectral variability without losing extracted spectral features. It can be presented as

$$\mathcal{L}_{\text{SAD}} = \frac{1}{N}\sum_{i=1}^{N}\arccos\left(\frac{\langle\hat{\mathbf{x}}_i, \hat{\mathbf{x}}_i\rangle}{\|\hat{\mathbf{x}}_i\|_2\|\hat{\mathbf{x}}_i\|_2}\right). \tag{32}$$

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The mse is defined as

$$\mathcal{L}_{\text{mse}} = \frac{1}{N}\sum_{i=1}^{N}\|\hat{\mathbf{x}}_i - \mathbf{x}_i\|_2^2 \tag{33}$$

where $\hat{\mathbf{x}}_i$ represents the ith pixel in reconstructed HSI, $\mathbf{x}_i$ represents the corresponding pixel in observed HSI, and N is the number of pixels. The total loss can be expressed by combining the loss of DM as follows:

$$\begin{aligned}\mathcal{L}_{\text{total}} &= \mathcal{L}_{\text{diff}} + \lambda\mathcal{L}_{\text{rec}} \\ &= \mathcal{L}_{\text{diff}} + \lambda_1\mathcal{L}_{\text{sad}} + \lambda_2\mathcal{L}_{\text{mse}}\end{aligned} \tag{34}$$

where λ_1 and λ_2 are tradeoff parameters that control the balance between the contribution of each loss term.

IV. Experiments and Results

In this section, we conduct experiments on one synthetic and three real HSI datasets to evaluate the performance of the proposed method. For comparison, we select four LMM-based methods and three NLMM-based methods. Among the LMM-based methods, we include one traditional method (VCA+FCLS [4], [7]) and three DL networks (UnDIP [34], CyCU-Net [33], and MSNet [37]). For the NLMM-based methods, we choose one traditional algorithm (NU-BGBM [16]) and two DL networks (DMBU [45] and DAAN [47]). For all the compared DL methods except UnDIP, the initial endmembers are extracted by VCA.

A. Evaluation Metrics

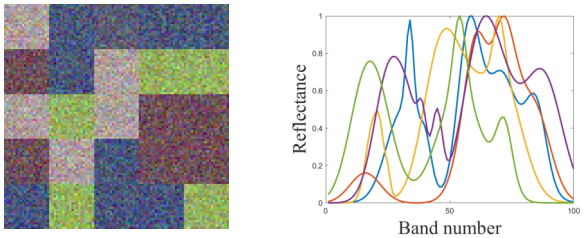
In this work, we introduce three commonly used evaluation metrics to assess the unmixing performance quantitatively, namely, SAD, root-mean-square error (RMSE), and reconstruction error (RE). Their specific expressions are given as follows:

$$\text{SAD}\left(\hat{\mathbf{e}}_j, \mathbf{e}_j\right) = \arccos\left(\frac{\hat{\mathbf{e}}_j^{\mathbf{T}}\cdot\mathbf{e}_j}{\|\hat{\mathbf{e}}_j\|\cdot\|\mathbf{e}_j\|}\right) \tag{35}$$

$$\text{RMSE}\left(\hat{\mathbf{A}}_j, \mathbf{A}_j\right) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\mathbf{A}}_{ji} - \mathbf{A}_{ji}\right\|_2^2} \tag{36}$$

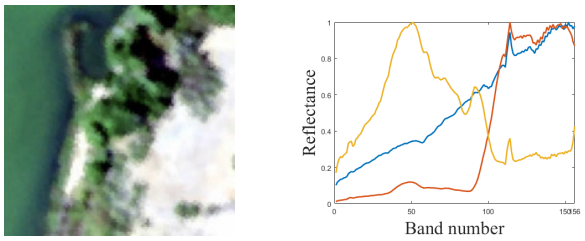
$$\text{RE}\left(\hat{\mathbf{X}}, \mathbf{X}\right) = \sqrt{\frac{1}{NL}\sum_{i=1}^{N}\sum_{r=1}^{L}\left\|\hat{\mathbf{X}}_{ir} - \mathbf{X}_{ir}\right\|_2^2} \tag{37}$$

where $\hat{\mathbf{e}}_j$ is the jth extracted endmember; $\mathbf{e}_j$ denotes the true endmember; $\hat{\mathbf{A}}_j$ and $\mathbf{A}_j$ represent the estimated abundance of jth endmember and the corresponding real abundance, respectively; and $\hat{\mathbf{X}}$ and $\mathbf{X}$ are the reconstructed and original HSI, respectively. Here, $\hat{\mathbf{X}}$ in the NLMM-based algorithm is computed through its nonlinear reconstruction formulation, $\mathbf{N}$ represents the number of pixels in his, and $\mathbf{L}$ is the number of bands. According to formulas, lower values of three metrics indicate better unmixing performances.



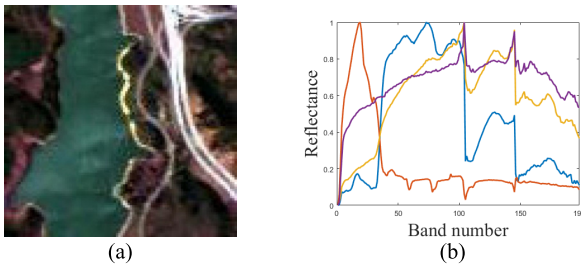
(a) (b)

Fig. 5. SD. (a) Simulated image. (b) Endmember spectra.



(a) (b)

Fig. 6. Samson dataset. (a) RGB image. (b) Endmember spectra.



(a) (b)

Fig. 7. Jasper Ridge dataset. (a) RGB image. (b) Endmember spectra.

B. Hyperspectral Datasets

1) Synthetic Dataset: The Synthetic dataset (SD) consists of five randomly selected endmembers with 100 spectral bands. The abundances of SD containing 100×100 pixels satisfy the ANC and ASC constraints. The simulated image is shown in Fig. 5(a), and the spectra of five endmembers are shown in Fig. 5(b). The dataset enables independent noise injection on both endmembers and generated images based on [60]. This design better simulates real-world noise conditions in endmembers and evaluates the unmixing algorithm's capability to extract pure endmembers from noisy observations. Quantitative measurements confirm signal-to-noise ratios (SNRs) of 6.4, 13.4, and 25.5 dB for the noise-contaminated data.

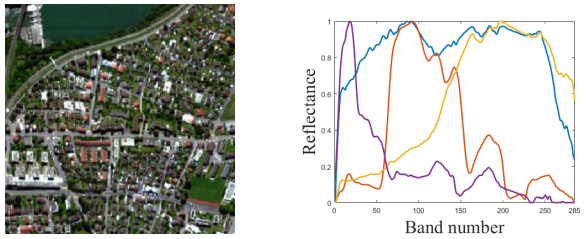
2) Samson Dataset: The Samson dataset was captured via the SAMSON sensor [61] and consists of 952×952 pixels and 156 spectral bands ranging from 401 to 889 nm. As shown in Fig. 6, we focus on a subimage measuring 95×95 pixels in this experiment. Three dominant endmembers are investigated: #1 Soil, #2 Tree, and #3 Water.

3) Jasper Ridge Dataset: The Jasper Ridge dataset was obtained by an Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) and consists of 512×614 pixels, as shown in Fig. 7. Each pixel records 224 spectral bands ranging from

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TABLE I
QUANTITATIVE COMPARISONS ON THE SD AT SNR =13.4 DB

Type		LMM-based				NLMM-based			
Method		V/F	UnDIP	CyCU	MSNet	NU-BGBM	DMBU	DAAN	DEAE
SAD	EM#1	0.0607	0.5700	0.0665	0.0673	0.0665	0.0092	0.0658	0.0083
	EM#2	0.1168	0.2507	0.1061	0.1098	0.1061	0.0215	0.1100	0.0139
	EM#3	0.0615	0.2424	0.0660	0.0822	0.0660	0.0331	0.0672	0.0242
	EM#4	0.0488	0.3442	0.0222	0.0526	0.0222	0.0063	0.0210	0.0110
	EM#5	0.1571	0.2538	0.1367	0.1548	0.1367	0.0091	0.1344	0.0140
mean SAD		0.0890	0.3322	0.0795	0.0933	0.0795	0.0158	0.0797	0.0143
RMSE	EM#1	0.0769	0.4472	0.3254	0.0589	0.1352	0.1061	0.0889	0.0015
	EM#2	0.0854	0.0485	0.2767	0.0496	0.1793	0.0924	0.0736	0.0015
	EM#3	0.0574	0.0886	0.3324	0.0552	0.1334	0.1023	0.0827	0.0024
	EM#4	0.0418	0.4495	0.3420	0.0330	0.0549	0.0799	0.0264	0.0007
	EM#5	0.0955	0.0886	0.1291	0.0643	0.1924	0.1009	0.0768	0.0020
mean RMSE		0.0714	0.2245	0.2811	0.0522	0.1390	0.0963	0.0697	0.0016
RE		0.0542	0.4213	0.2839	0.0532	0.1275	0.0467	0.0488	0.0065



(a) (b)

Fig. 8. Apex dataset. (a) RGB image. (b) Endmember spectra.

380 to 2500 nm. For our experiment, we focus on a subimage measuring 100×100 pixels. We retain a total of 198 bands after removing the bands affected by high levels of water vapor and atmospheric interference. Four endmembers are manually selected from the Jasper Ridge dataset: #1 Soil, #2 Tree, #3 Water, and #4 Road.

4) Apex Dataset: The Apex dataset, acquired by the APEX sensor [62], contains an image of 1500×1000 pixels with 285 spectral bands covering 413–2400 nm, as shown in Fig. 8. For our experiment, a subimage of size 110×110 pixels is selected for evaluation. Four endmembers are identified: #1 Soil, #2 Tree, #3 Water, and #4 Road.

C. Parameter Setting

We used the AdamW optimizer on four datasets: the SD, the Samson dataset, the Jasper dataset, and the Apex dataset, with global learning rates of 0.001, 0.01, 0.005, and 0.01, respectively. The weight decay of optimizer is set to 0.001. For the Diff-Decoder, we assigned distinct learning rates of 0.0001, 0.05, 0.0005, and 0.0001 to the four datasets. The length of the diffusion process T was set to 1000. The tradeoff parameters λ_1 and λ_2 introduced in (34) were both set to 0.001. To evaluate the effects of λ_1 and λ_2, we conducted experiments on the Jasper dataset using three metrics, with the corresponding unmixing results visualized in Fig. 9.

Due to the significant magnitude disparity between the loss functions of the two models, the AE's loss would severely interfere with the DM's training during joint optimization. To address this, we implement a sequential training strategy where the linear AE is first pretrained exclusively for 800 epochs to establish stable representations. Subsequently, the entire DEAE, including the linear AE and Diff-Decoder, is jointly trained for another 800 epochs, allowing the diffusion process to adapt to the learned latent space while maintaining feature stability through shared optimization. The training loss curves for linear AE and entire DEAE are shown in Fig. 10. The hyperparameters of the other methods we compared in this article have been adjusted to optimize their performance.

D. Experiments on SD and Analysis

1) Unmixing Performance: The quantitative results of RMSE, SAD, and RE for each endmember on SD, obtained using different algorithms, are presented in Table I, with the SNR set to 13.4 dB. The extracted endmembers and the corresponding abundance maps are shown in Figs. 11 and 12. The best and second-best results are highlighted in red and blue, respectively. In SD, DEAE achieves the best mean SAD, mean RMSE, and RE, securing either the best or second-best performance in each endmember. This demonstrates its superior feature extraction capability, better noise resistance, and powerful signal reconstruction ability from the diffusion-enhanced AE architecture. For endmember extraction, DMBU achieves the second-best results, effectively extracting endmember features by separately estimating the linear and nonlinear components through multitask learning. Moreover, NLMM-based algorithms generally outperform LMM-based algorithms in RE by 20%–40%, owing to the superior data-fitting capability provided by their higher order spectral mixing models.

2) Noise Robustness Analysis: To investigate the robustness of the proposed method, noise with different SNRs from 6.4 to 25.5 dB is added in the synthetic experiment. Correspondingly, the quantitative experimental results are presented in Table II. As shown in Fig. 13, the unmixing accuracy of these methods tends to promote as the noise decreases. The line in blue represents DEAE, indicating that our proposed method

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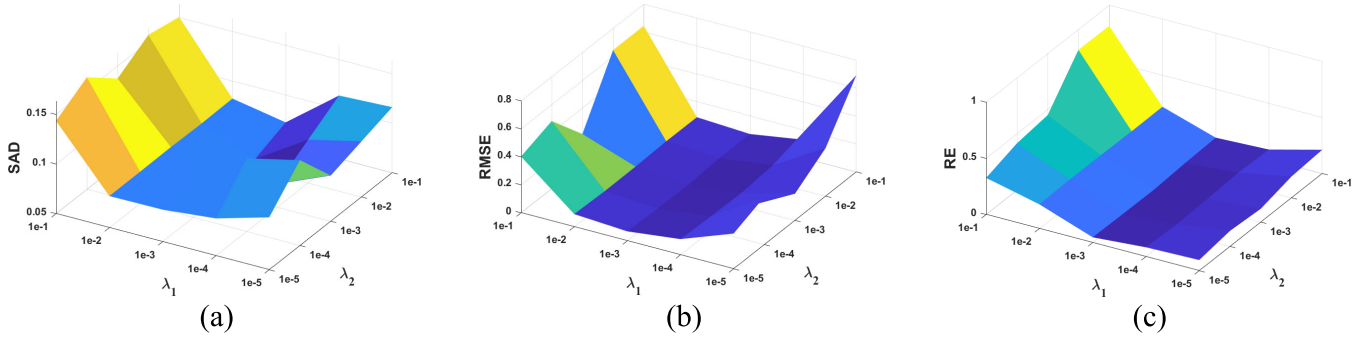


Fig. 9. Unmixing accuracies under different hyperparameters λ_1 and λ_2. (a) SAD. (b) RMSE. (c) RE.

TABLE II
QUANTITATIVE RESULTS OF MEAN RMSE, MEAN SAD, AND RE FOR THE SD AT DIFFERENT SNRS

Type		LMM-based				NLMM-based			
Method		V/F	UnDIP	CyCU	MSNet	NU-BGBM	DMBU	DAAN	DEAE
mean SAD	6.4dB	0.1807	0.3407	0.1664	0.1893	0.1664	0.1091	0.1647	0.0906
	13.4dB	0.0890	0.3322	0.0795	0.0933	0.0795	0.0158	0.0797	0.0143
	25.5dB	0.0359	0.3253	0.0271	0.0264	0.0271	0.0178	0.0279	0.0027
mean RMSE	6.4dB	0.1351	0.2662	0.2542	0.0935	0.2663	0.1643	0.1459	0.0644
	13.4dB	0.0714	0.2245	0.2811	0.0522	0.1390	0.0963	0.0697	0.0016
	25.5dB	0.0208	0.1871	0.2751	0.0246	0.0400	0.0922	0.0828	0.0006
RE	6.4dB	0.1159	0.3966	0.3103	0.1041	0.2218	0.0947	0.1459	0.0557
	13.4dB	0.0542	0.4213	0.2839	0.0532	0.1275	0.0467	0.0488	0.0065
	25.5dB	0.0222	0.2334	0.2651	0.0172	0.0383	0.0518	0.0459	0.0013

TABLE III
QUANTITATIVE RESULTS OF MEAN RMSE, MEAN SAD, AND RE FOR THE SAMSON DATASET

Type		LMM-based				NLMM-based			
Method		V/F	UnDIP	CyCU	MSNet	NU-BGBM	DMBU	DAAN	DEAE
SAD	Soil	0.0232	0.0501	0.0361	0.0192	0.0232	0.0219	0.0230	0.0091
	Tree	0.0330	0.0183	0.0355	0.0283	0.0330	0.0249	0.0452	0.0308
	Water	0.1282	0.1092	0.1898	0.0648	0.1282	0.6928	0.1553	0.0275
mean SAD		0.0615	0.0592	0.0871	0.0375	0.0615	0.2465	0.0745	0.0225
RMSE	Soil	0.2575	0.2670	0.1828	0.0581	0.2181	0.1884	0.1344	0.0494
	Tree	0.1996	0.2520	0.1921	0.0532	0.2318	0.1990	0.0831	0.0406
	Water	0.3673	0.4261	0.1578	0.0247	0.0888	0.3682	0.1351	0.0235
mean RMSE		0.2748	0.3150	0.1776	0.0453	0.1796	0.2519	0.1175	0.0378
RE		0.2295	0.2602	0.2810	0.0212	0.2024	0.2008	0.0892	0.0189

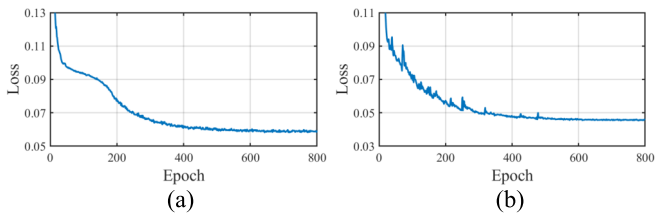


Fig. 10. Sequential training loss curves on the Samson dataset. (a) Linear AE training. (b) DEAE joint training.

exhibits strong robustness against noise. This superior performance can be attributed to the diffusion-enhanced decoder architecture, which provides exceptional data-fitting capability that accurately models complex spectral mixtures and powerful latent variable extraction that effectively estimates the transition probabilities **P** while suppressing noise components. Traditional algorithms generally perform well in unmixing under low-noise conditions, suggesting that they still hold certain advantages in scenarios with simpler ground cover conditions. In low noise conditions, the algorithm-based NLMMs do not show a significant advantage over the algorithms based on LMM. This is because the nonlinear characteristics of the data are weaker under the high SNR, while the linear features are more dominant, enabling linear-based algorithms to better simulate light reflection among ground covers. Nevertheless, the proposed method exhibits robust unmixing performance across different noise levels.

E. Experiments on Real Datasets

1) Samson Dataset: The quantitative results, the extracted endmembers, and the corresponding abundance maps of Samson dataset are shown in Table III, Figs. 14 and 15, respec-

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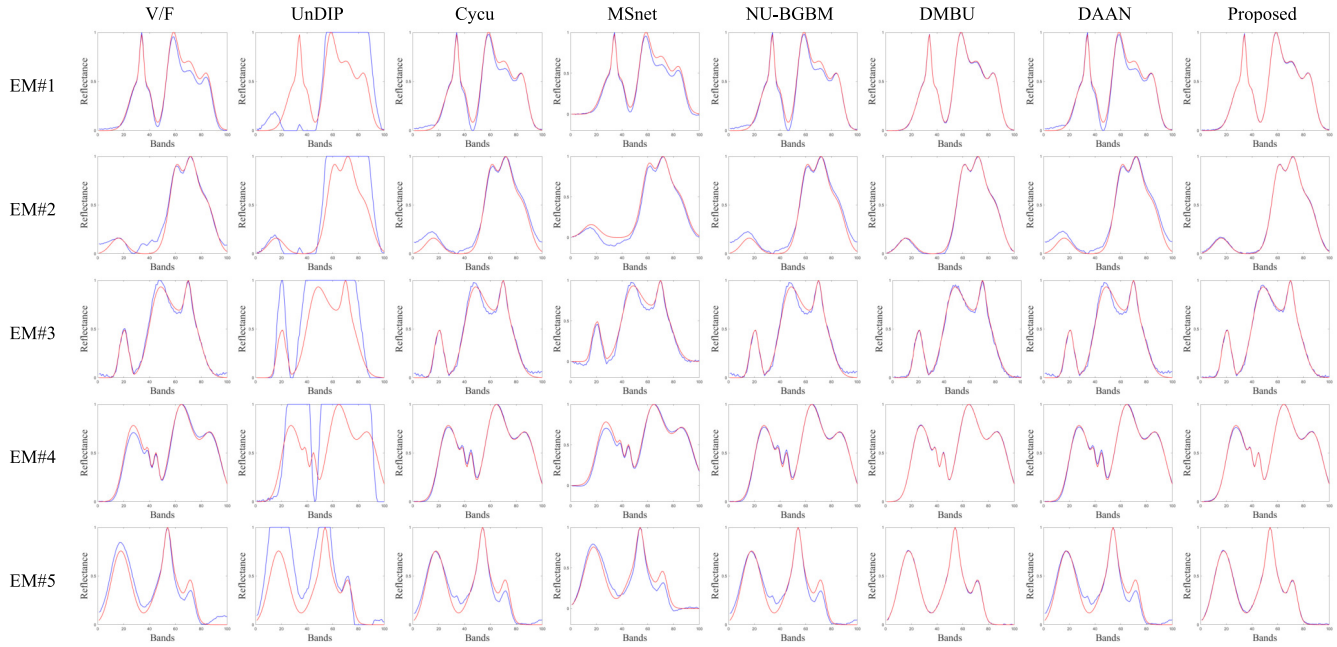


Fig. 11. Estimated endmembers by different methods (blue curves) and the corresponding ground truth (red curves) on SD.

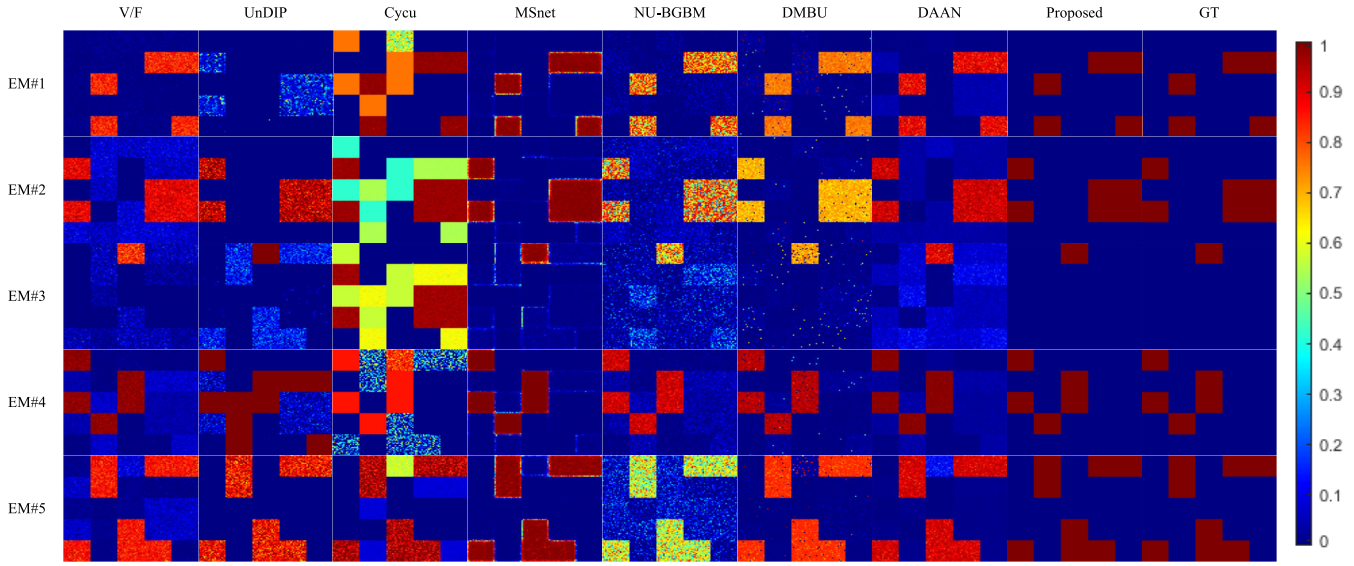


Fig. 12. Estimated abundance maps for different endmembers on SD.

tively. Samson contains only three endmembers with relatively simple combinations, enabling high accuracy in endmember extraction across all algorithms. Moreover, due to simpler ground cover conditions, higher order interactions are less frequent, while lower order interactions dominate during the propagation of light, allowing the algorithms based on LMM to perform effectively. Among these methods, MSNet, through its multistage AE structure, effectively extracted features from the observed spectra, achieving second-best unmixing results. Meanwhile, DEAE obtains the best RMSE, SAD, and RE. DEAE demonstrates exceptional performance in endmember extraction, outperforming the second-best algorithm by 40%. This significant improvement indicates that even in relatively simple scenarios, the EMLM can achieve precise extraction of endmember characteristics.

In abundance estimation results shown in Fig. 15, NU-BGBM and DMBU show significant deviations in tree estimation, with DMBU misclassifying part of the "Tree" as "Water." Both NU-BGBM and DMBU are based on the bilinear spectral mixing model, which considers only the quadratic terms in higher order scattering. As a result, inaccuracies in estimating the bilinear component significantly impact the unmixing accuracy. DAAN estimates an abundance map closely matching the ground truth. However, errors in water endmember estimation lead to some misclassifications in the right region of its water abundance map. The abundance

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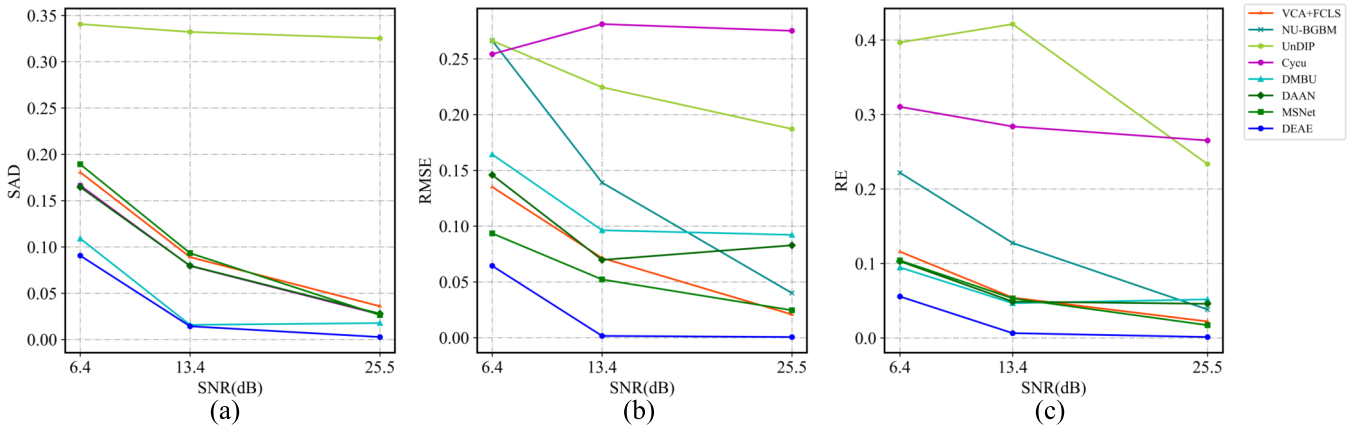


Fig. 13. Robustness evaluation of the SD with different SNR values. (a) SAD. (b) RMSE. (c) RE.

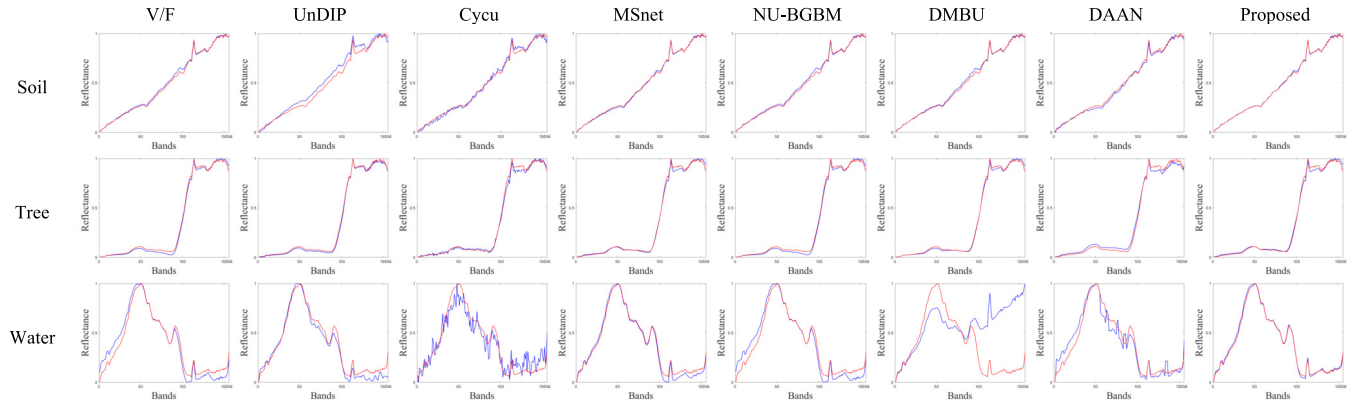


Fig. 14. Estimated endmembers by different methods (blue curves) and the corresponding ground truth (red curves) on the Samson dataset.

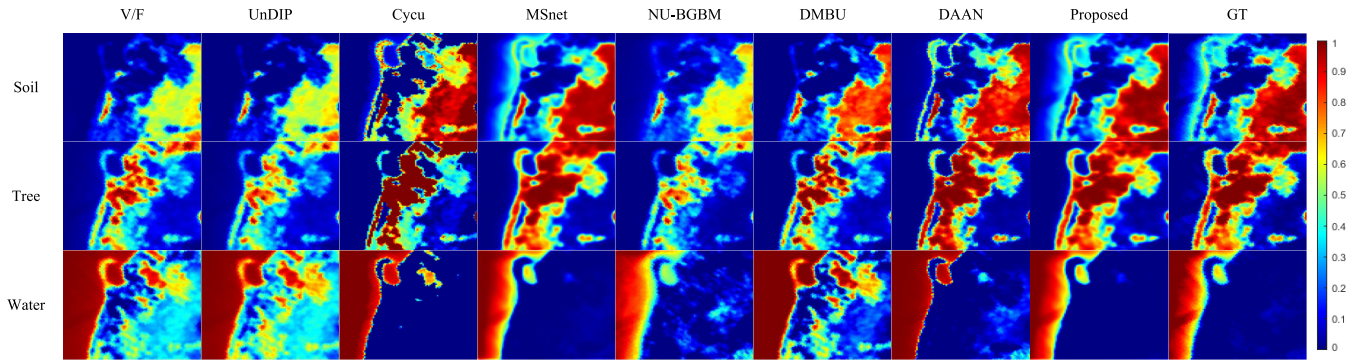


Fig. 15. Estimated abundance maps for different endmembers on the Samson dataset.

maps produced by DEAE are the most similar to the ground truth. Built on EMLM that accounts for scattering of infinite orders, it achieves smoother transitions at the boundaries between different endmembers and delivers precise abundance estimates.

2) Jasper Ridge Dataset: Table IV shows the quantitative results of the RMSE, SAD, and RE on the Jasper Ridge dataset by the different methods. The extracted endmember and the corresponding abundance maps are shown in Figs. 16 and 17, respectively. The distribution of ground covers in Jasper Ridge is more complex, with tight combinations of "Road," "Tree," and "Soil," imposing higher demands on the unmixing method performance. As shown in Table IV, DMBU and CYCU achieved notable unmixing performances in endmember estimation. DMBU performed optimally or near optimally on "Soil," "Road," and "Water," highlighting the advantages of algorithms based on NLMM in endmember estimation under complex ground conditions. CyCU-Net, by cascading two AEs to continuously extract features from the input HSI, also performed well on the Jasper Ridge dataset. Both DEAE and MSNet demonstrate superior performance in endmember extraction compared to other algorithms, with

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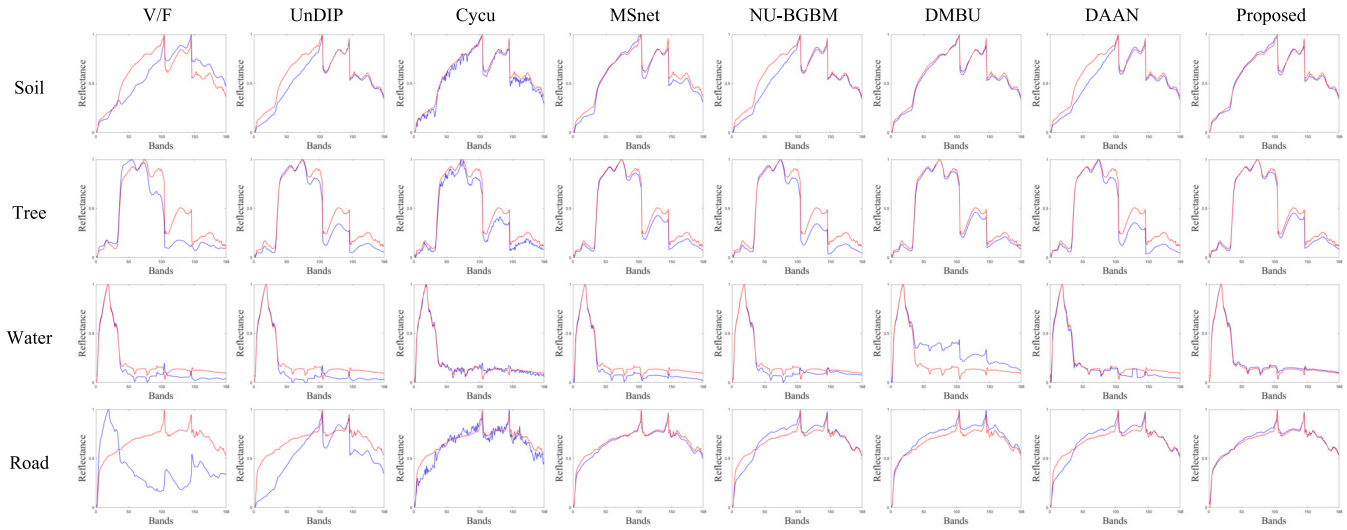


Fig. 16. Estimated endmembers by different methods (blue curves) and the corresponding ground truth (red curves) on the Jasper Ridge dataset.

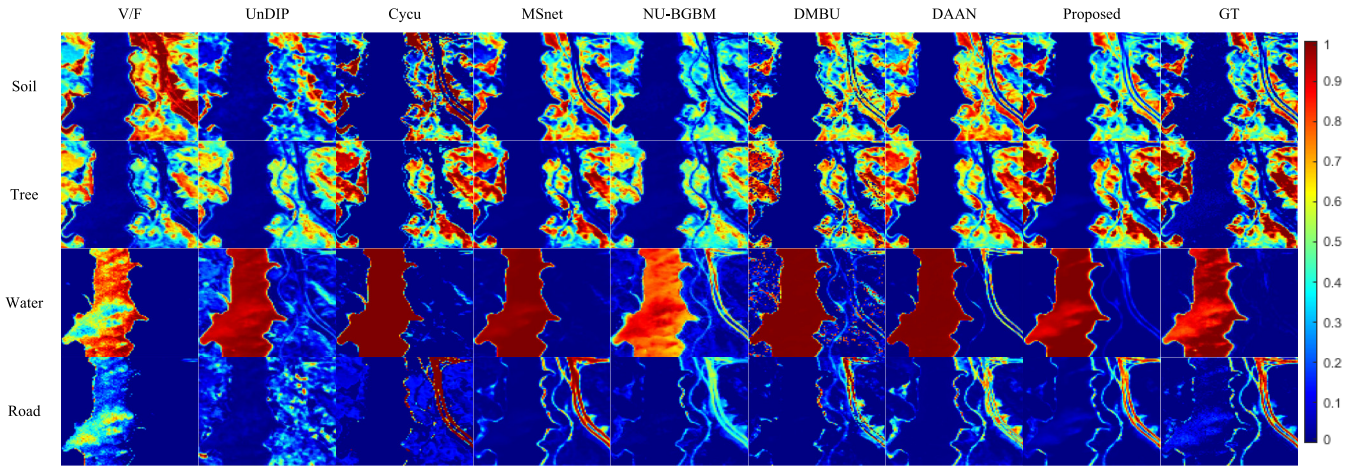


Fig. 17. Estimated abundance maps for different endmembers on the Jasper Ridge dataset.

TABLE IV
QUANTITATIVE RESULTS OF MEAN RMSE, MEAN SAD, AND RE FOR THE JASPER RIDGE DATASET

Type		LMM-based				NLMM-based			
Method		V/F	UnDIP	CyCU	MSNet	NU-BGBM	DMBU	DAAN	DEAE
SAD	Soil	0.2857	0.1535	0.0858	0.0791	0.1538	0.0487	0.1446	0.0526
	Tree	0.2138	0.2522	0.0772	0.0286	0.1395	0.3739	0.1328	0.0468
	Water	0.2254	0.1149	0.0539	0.0526	0.1180	0.0271	0.1203	0.0332
	Road	0.6105	0.2626	0.1013	0.0246	0.0839	0.0273	0.0818	0.0240
mean SAD		0.3338	0.1958	0.0796	0.0462	0.1238	0.1193	0.1199	0.0392
RMSE	Soil	0.1744	0.1595	0.0812	0.0920	0.1711	0.1685	0.0943	0.0498
	Tree	0.1516	0.1438	0.1002	0.0707	0.2110	0.2033	0.1321	0.0460
	Water	0.2824	0.2161	0.1248	0.1087	0.1444	0.1154	0.1459	0.0677
	Road	0.2629	0.2217	0.1353	0.0854	0.1182	0.1408	0.1322	0.0771
mean RMSE		0.2178	0.1853	0.1104	0.0892	0.1612	0.1570	0.1261	0.0602
RE		0.1124	0.3716	0.3709	0.0457	0.1445	0.1177	0.0919	0.0380

DEAE achieving a 33% improvement over the second-best method in abundance estimation metrics.

As shown in Fig. 7, due to the high similarity between the spectral curves of "Water" and "Road" in the Jasper Ridge dataset, VCA+FCLS and UnDIP fail to distinguish between these two endmembers. CyCU accurately identifies all four endmembers but produces abundance maps with significant noise and blurred edge contours. This is because CyCU

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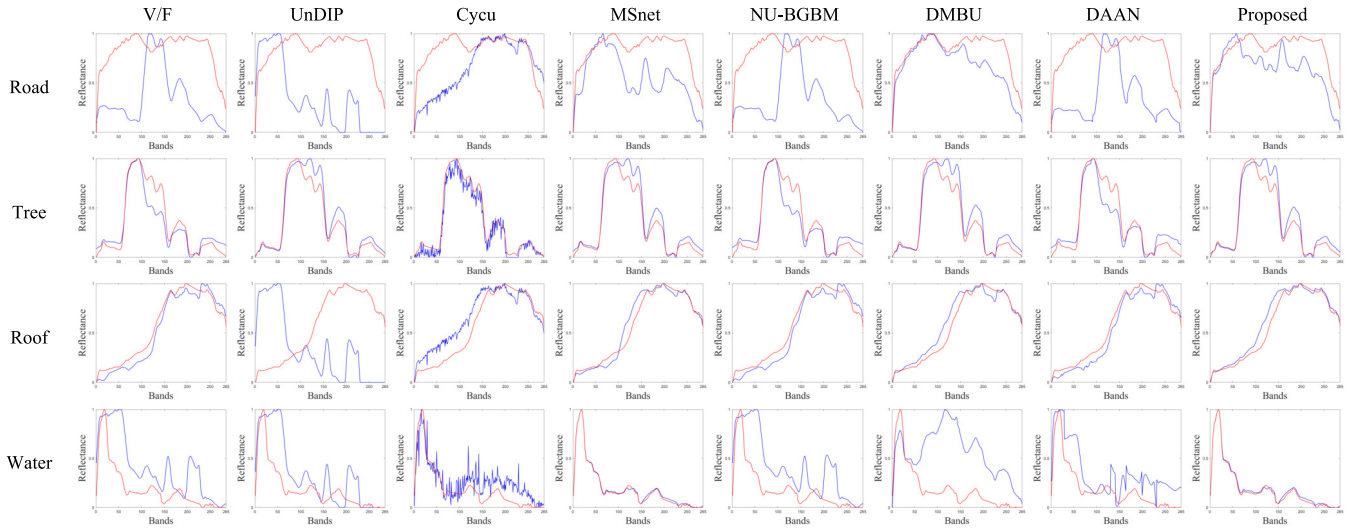


Fig. 18. Estimated endmembers by different methods (blue curves) and the corresponding ground truth (red curves) on the Apex dataset.

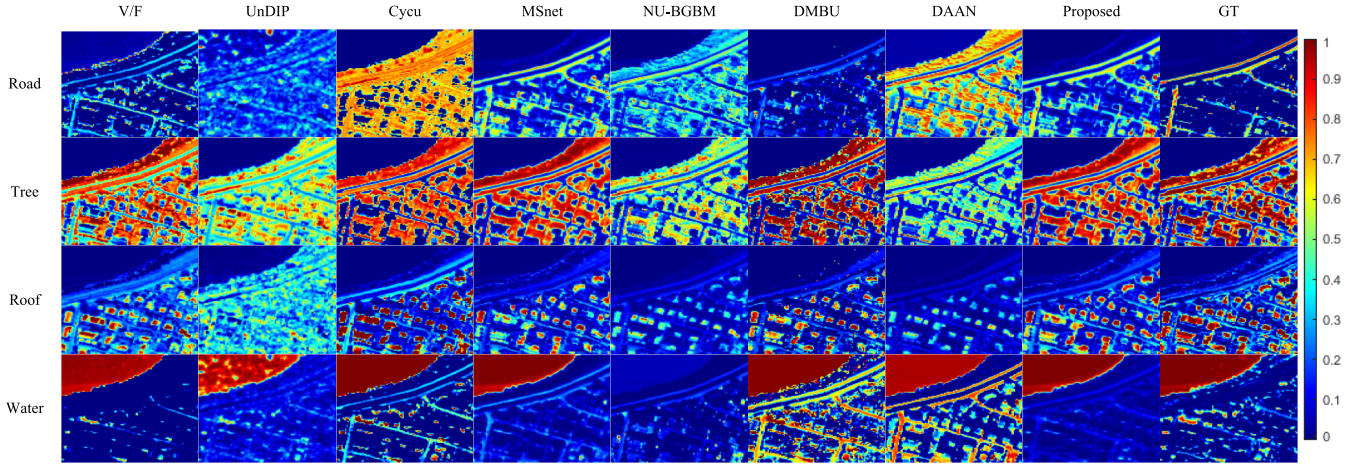


Fig. 19. Estimated abundance maps for different endmembers on the Apex dataset.

primarily relies on 1×1 convolutions, which are limited in extracting spatial features. MSNet exhibits minimal noise points and rich detail, achieving the best abundance estimation performance among LMM-based algorithms. Among NLMM-based algorithms, NU-BGBM delivers smooth edge contours and minor noise points but lacks detailed information and fails to distinguish between "Water" and "Road." DMBU provides richer detail but is heavily affected by noise, resulting in poor visual quality. DAAN produces abundance maps with detailed features and sharp edge contours that are similar to the actual ground objects. However, it also fails to effectively separate "Water" and "Road," compromising the unmixing accuracy. DEAE effectively captures the detailed features of all four complex endmembers, delivering the best visual performance.

3) Apex Dataset: The quantitative results on the Apex dataset are shown in Table V. The endmembers extracted by different methods and the corresponding visualized abundance maps are illustrated in Figs. 18 and 19, respectively. In the Apex dataset, except for the relatively independent water, the other three ground covers are closely mixed, presenting considerable difficulties for the unmixing task. Among them, the distribution of "Road" is relatively sparse, making it difficult to extract features. Except for MSNet, LMM-based methods cannot extract accurate endmember curves for "Road." DMBU and DEAE, which achieve the best SAD, extract the specific features of "Roads" relatively accurately, demonstrating the advantage of NLMM in highly mixed scenarios. Nevertheless, DMBU has a poor unmixing effect on water, confusing water with tree in both endmember extraction and abundance estimation, thereby revealing the limitations of the BMM model. The proposed method achieves the best results in all three metrics. The method demonstrates particularly superior performance in RE, outperforming the second-best algorithm by 45%. This significant advantage strongly validates that the EMLM-based unmixing method can more accurately simulate and reconstruct real-world ground features in complex environments with urban characteristics.

In the abundance maps, VCA+FCLS achieves favorable abundance estimation results on the Apex dataset, accurately

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TABLE V
QUANTITATIVE RESULTS OF MEAN RMSE, MEAN SAD, AND RE FOR THE APEX DATASET

Type		LMM-based				NLMM-based			
Method		V/F	UnDIP	CyCU	MSNet	NU-BGBM	DMBU	DAAN	DEAE
SAD	Road	0.6466	0.8182	0.7587	0.2917	0.6415	0.1845	0.6264	0.2325
	Tree	0.2193	0.1359	0.1132	0.1327	0.2222	0.1508	0.2161	0.1315
	Roof	0.1091	1.2193	0.1923	0.1092	0.1066	0.1164	0.1003	0.0791
	Water	0.4997	0.4854	0.4112	0.0460	0.5092	0.8182	0.4032	0.0450
mean SAD		0.3687	0.6647	0.3689	0.1449	0.3699	0.3175	0.3365	0.1220
RMSE	Road	0.2064	0.2086	0.2171	0.1843	0.7669	0.1892	0.4366	0.1658
	Tree	0.1435	0.2560	0.1823	0.1330	0.2264	0.1622	0.3241	0.1235
	Roof	0.1342	0.2781	0.1122	0.1541	0.1440	0.1449	0.2725	0.1306
	Water	0.1336	0.1312	0.1166	0.1528	1.8808	0.2954	0.3067	0.1364
mean RMSE		0.1544	0.2185	0.1570	0.1561	0.7545	0.1979	0.3350	0.1391
RE		0.1947	0.3347	0.2979	0.1384	1.0323	0.2517	0.2371	0.0757

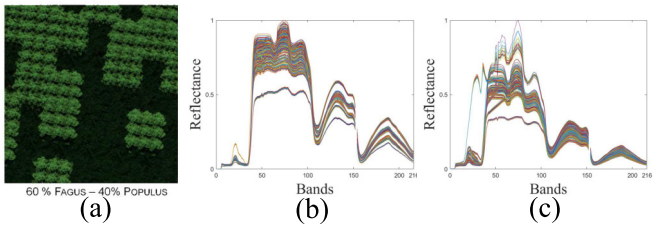


Fig. 20. Ray Tracing dataset. (a) High-resolution image of the forest with 60% beeches and 40% poplars. (b) Beech's spectral curves in each pixel. (c) Poplar's spectral curves in each pixel.

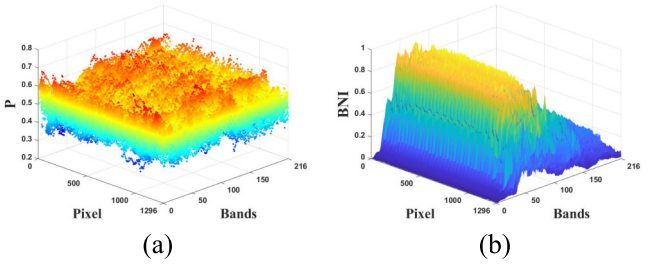


Fig. 21. Analysis of the nonlinear parameter **P**. (a) Calculated **P** vectors of each pixel by DEAE. (b) BNI vectors of each pixel.

identifying all four endmembers and estimating abundance maps closely similar to the real ground covers, demonstrating the stability of traditional algorithms. UnDIP gets a low RMSE. However, its abundance maps are heavily affected by noise, resulting in poor visual quality. CyCU delivers the best abundance estimation results for "Roof" and "Water," effectively capturing the distribution and contour of these two ground covers. However, its performance on "Road" is inferior, failing to accurately represent the edge contour. DMBU also achieves superior abundance estimation results with accurate endmember identification and relatively clear details. Nonetheless, its abundance maps are marred by significant noise, affecting visual perception. DEAE achieves optimal abundance estimation results, with its abundance maps exhibiting the most detailed visual representation. As demonstrated, DEAE not only accurately identifies "Roof" features but also preserves rich structural details, highlighting the positive impact of DMs' superior generative capability on unmixing performance.

F. Analysis

1) Nonlinear Effect Analysis on Ray Tracing Datasets: To further demonstrate the effectiveness of DEAE, we conducted a series of experiments on ray tracing datasets [63], which is a physically based synthetic HSI generated using the PBRT framework [64]. The scene in PBRT is constructed by specifying several distinct components, each responsible for handling components like light emitters, imaging devices, ground covers, integration methods, and object geometry. The dataset comprises two endmembers (#1: beech and #2: poplar) across 36×12 pixels, totaling 216 spectral bands. In this dataset, poplar initially accounts for 80% of the mixture and is gradually replaced by beech in increments of 20%. As a result, the dataset includes four distinct land cover distributions, labeled Forest_b20p80, Forest_b40p60, Forest_b60p40, and Forest_b80p20. The high-resolution detail of Forest_b60p40 is illustrated in Fig. 20(a). Notably, each pixel contains a unique combination of endmember characteristics, resulting in pixel-specific endmember spectra. Therefore, each endmember matrix comprises 36×12 individual spectral curves, representing the variation at each pixel location, as shown in Fig. 20(b) and (c).

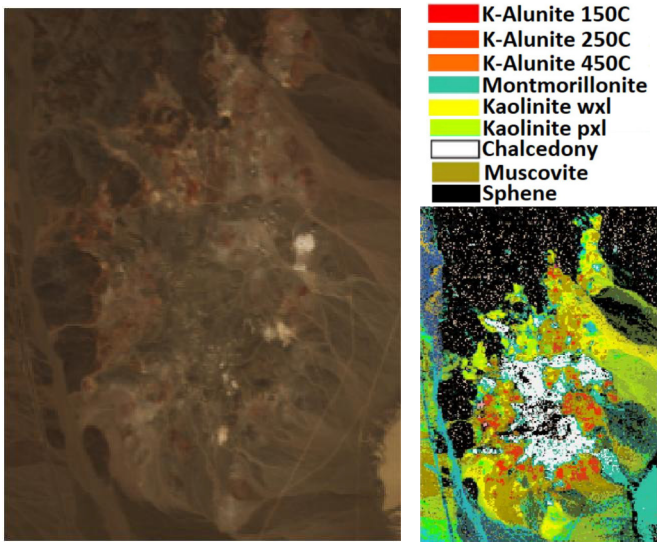
The quantitative results are presented in Table VI, with the optimal and suboptimal results highlighted in red and blue, respectively. A significant challenge in evaluating endmember estimation accuracy arises from the inherent pixel-wise variability of endmembers in ray tracing datasets, which contrasts with most DL approaches designed for whole-image endmember estimation. Consequently, we restrict our evaluation to abundance estimation and overall reconstruction accuracy using RMSE and RE metrics. DEAE demonstrated consistently strong performance across both metrics.

Further analysis of transition probabilities **P** reveal several important characteristics, as shown in Fig. 21(a). **P** is rigorously constrained within the interval (0, 1) and exhibits two fundamental independence properties. First, pixel-wise independence emerges from the heterogeneous ground covers across different pixels, where varying material proportions generate distinct nonlinear effects. Second, band-wise

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TABLE VI
QUANTITATIVE RESULTS OF RMSE AND RE FOR RAY TRACING DATASETS

Method		V/F	UnDIP	Cycu	Msnet	NU_GBM	DMBU	DAAN	DEAE
RMSE	Forest_b20p80	0.4617	0.1216	0.2668	0.3443	0.8414	0.1382	0.2494	0.0350
	Forest_b40p60	0.4534	0.1899	0.2494	0.2958	0.6038	0.0875	0.1682	0.0531
	Forest_b60p40	0.4442	0.0995	0.1858	0.1298	0.3857	0.1427	0.1481	0.0563
	Forest_b80p20	0.2305	0.0898	0.1653	0.2454	0.5749	0.0845	0.0925	0.0159
RE	Forest_b20p80	0.4409	0.0865	0.1290	0.0991	0.2080	0.3670	0.0657	0.0586
	Forest_b40p60	0.4540	0.0793	0.1298	0.0907	0.1985	0.0385	0.0565	0.0516
	Forest_b60p40	0.4665	0.0727	0.1338	0.0863	0.2078	0.0225	0.0507	0.0437
	Forest_b80p20	0.1328	0.1194	0.1404	0.0937	0.2479	0.0476	0.0428	0.0411



(a) (b)

Fig. 22. Cuprite dataset. (a) RGB image. (b) Ground truth mineral map.

independence stems from wavelength-dependent nonlinear interactions, a phenomenon well-documented in previous studies [21], [65]. We introduce a band-wise nonlinear intensity (BNI) metric to quantitatively evaluate the intensity of nonlinear interaction, which is defined as

$$\mathrm{BNI} = \frac{\mathbf{X} - \mathbf{EA}}{\|\mathbf{X}\|_F} \tag{38}$$

where the division is performed element-wise and $\mathbf{X}, \mathbf{E}$, and $\mathbf{A}$ represent the observed HSI, endmember matrix, and abundance matrix, respectively. Higher values indicate stronger deviation from linear representation. The visualization of BNI in Fig. 21(b) demonstrates that after removing the influence of linear components, the nonlinear variations between pixels are significantly reduced, while the band-dependent nonlinear effects exhibit distinct patterns. Notably enhanced BNI values are observed in the spectral range of bands 50–100, showing a measurable correlation with the spectral peaks of endmember curves.

2) Visualization Analysis on Cuprite Dataset: The AVIRIS Cuprite dataset, collected over a mineral-rich region in Nevada, USA, has been widely used as a benchmark for visualizing unmixing results. As shown in Fig. 22, a subimage of size 250 × 191 pixels with 188 spectral bands is used in the experiment. The visualization results in Fig. 23 demonstrate that the proposed DEAE method achieves accurate abundance estimation while effectively preserving spatial details across diverse ground covers.

TABLE VII
ABLATION EXPERIMENTS ON THE JASPER RIDGE DATASET

Component			Metrics		
SOCA	Diff-Decoder	$\mathcal{L}_{diff}$	SAD	RMSE	RE
✓	✗	✗	0.0736	0.0854	0.0968
✓	✓	✗	0.0409	0.0665	0.0617
✗	✓	✓	0.0523	0.0713	0.0497
✓	✓	✓	**0.0392**	**0.0602**	**0.0380**

Fig. 24 presents the visualization of transition probabilities $\mathbf{P}$. Specifically, Fig. 24(a) displays a grayscale image of a randomly selected band from the 188 bands of $\mathbf{P}$, where significant variations in transition probabilities can be observed among key ground covers. Fig. 24(b) illustrates the strict nonnegativity and concentrated distribution of probabilities in $\mathbf{P}$. To enhance visualization clarity through energy compaction, principal component analysis (PCA) is applied to $\mathbf{P}$, and the first three principal components are shown in Fig. 24(c)–(e).

G. Ablation Experiments

To evaluate the effectiveness of each component, we conduct an ablation study on the Jasper Ridge dataset. Specifically, we examined the impact of the SOCA module, the Diff-Decoder, and its associated loss function $\mathcal{L}_{\text{diff}}$. The quantitative results are presented in Table VII. The experimental results indicate that both the SOCA module and the Diff-Decoder play a critical role in endmember extraction and abundance estimation. Furthermore, the introduction of $\mathcal{L}_{\text{diff}}$ significantly improves the reconstruction accuracy of the HSI. This demonstrates the necessity of incorporating all three components for optimal unmixing performance.

H. Computational Cost

We evaluated the computational cost of the proposed method and other compared methods on the SD with 13.4-dB noise. The results are shown in Table VIII. All methods were implemented on a computer with an Intel Core i9-14900K CPU and a NVIDIA RTX 4090 GPU as the hardware platform.

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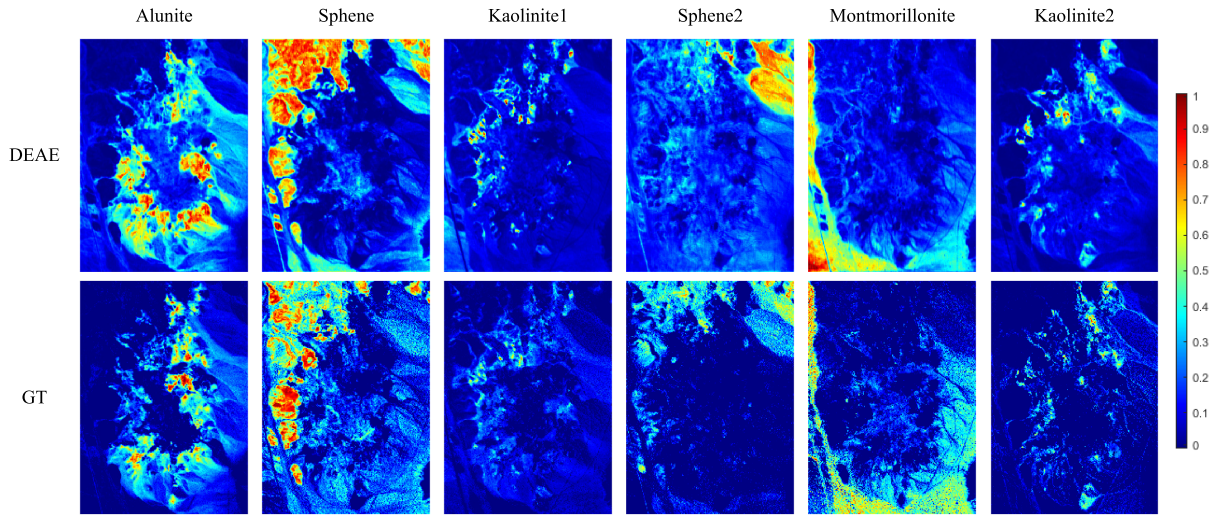


Fig. 23. Estimated abundance maps for different endmembers on the Cuprite dataset.

TABLE VIII
COMPUTATIONAL COST OF DIFFERENT METHODS ON SD AT SNR = 13.4 DB

Type	LMM-based				NLMM-based			
Method	VCA+FCLS	UnDIP	CyCU-Net	MSNet	NU-BGBM	DMBU	DAAN	DEAE
Time(s)	1.48	3176.02	4.56	8.46	21.27	58.08	26.22	77.2208
Params	/	1281089	22903	460080	/	151500	1096219	3931809

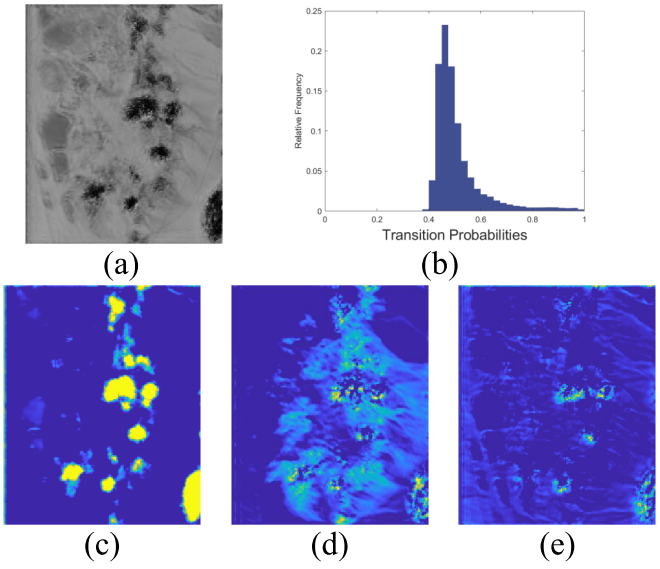


Fig. 24. Visualization of transition probabilities **P**. (a) Grayscale image of band 86. (b) Histogram of **P**. (c)–(e) First three principal components of **P**.

The VCA+FCLS, NU-BGBM, CyCU-Net, and DMBU methods were executed only on CPU, while the UnDIP, DAAN, MSNet, and DEAE methods were run on both CPU and GPU.

Since algorithms based on EMLM need to account for higher order interactions between light and endmembers during transition, DEAE requires more computation time compared to LMM-based algorithms and other NLMM-based algorithms. The large number of parameters is primarily due to the estimation of the transition probabilities **P** for each spectral band during network iterations. Furthermore, the incorporation of DMs significantly increases the model's parameter counts. While DEAE inevitably exhibits higher computational time and parameter counts, its overall computational overhead remains within acceptable limits. Moreover, leveraging the robust processing capabilities of modern GPUs, it has become practical to employ more sophisticated neural network architectures to enhance spectral unmixing accuracy.

V. CONCLUSION

In this article, we propose a DEAE network for unsupervised nonlinear HU, named DEAE. We introduce an RSCA to assign greater weights to spectral bands rich in information, effectively reducing redundancy. Additionally, we combine the powerful data-fitting ability of DL and design a diffusion-enhanced decoder structure to obtain more accurate transition probabilities **P**. Experiments on synthetic and real hyperspectral datasets validate the effectiveness and robustness of DEAE compared with other state-of-the-art unmixing approaches. In future work, we aim to explore network structures that more effectively integrate nonlinear mixing models with DL.

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